

Magnetized white dwarfs with mixed poloidal-toroidal fields

Draft – a certified $2 M_{\odot}$ configuration

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Draft July 31, 2026

ABSTRACT

Context. A white dwarf well above the Chandrasekhar mass requires support beyond electron degeneracy. A strong internal magnetic field can supply it, but a purely toroidal field has no exterior structure and so no dipole for magnetic braking to act on.

Aims. We construct a $2 M_{\odot}$ configuration carrying both components, and ask what a poloidal component costs when it is added to a toroidal-dominated equilibrium.

Methods. The toroidal field is solved self-consistently with the stellar structure by a self-consistent field iteration under a cold degenerate equation of state. A single solve carrying both components is not available in barotropy, where the toroidal field is confined to the closed-poloidal-line region and the reachable ratio is a few percent, so the poloidal component is solved on the converged density and imposed. Every configuration is gated on the virial error, on the star fitting inside its domain, and on the central density staying below the inverse-beta-decay threshold.

Results. The configuration reaches $M = 2.007 M_{\odot}$ at $\rho_c = 10^9 \text{ g cm}^{-3}$ with $E_{\text{tor}}/|W| = 0.203$, a virial error of 9.9×10^{-5} , and a mass converging monotonically under mesh refinement. The field is strongly toroidal-dominated, $B_t/B_p = 884$ in energy, with a peak of $4.28 \times 10^{13} \text{ G}$ and an exterior dipole of $5.9 \times 10^{10} \text{ G}$. The virial error of the pair tracks the imposed poloidal energy almost one for one, so a more balanced field leaves the certified regime: at $B_t/B_p = 8.8$ the virial error is twenty times the acceptance threshold.

Conclusions. A $2 M_{\odot}$ white dwarf with a strong toroidal field and an exterior dipole is constructible and certifiable, at the cost of a poloidal component carrying a tenth of a percent of the magnetic energy. It does not survive dynamically: evolved in three-dimensional magnetohydrodynamics it runs away in central density after about three dynamical times, at a rate insensitive to resolution and to the field strength, because the material that collapses is the axial column on which the toroidal field vanishes.

Key words. white dwarfs – stars: magnetic field – magnetohydrodynamics (MHD) – methods: numerical

1. Introduction

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The specific enthalpy has a closed form,

$$H(x) = \frac{8A}{B} \left[\sqrt{1+x^2} - 1 \right], \quad (3)$$

2. Equation of state

The background stellar structure throughout this work is that of a fully degenerate, zero-temperature electron gas (ztwd; Chandrasekhar 1935). Pressure and density are both parametrized by the normalized Fermi momentum $x \equiv p_F/(m_e c)$,

$$P(x) = A \left[x(2x^2 - 3) \sqrt{x^2 + 1} + 3 \sinh^{-1} x \right], \quad (1)$$

$$\rho(x) = B x^3, \quad (2)$$

where A is a fixed combination of fundamental constants (m_e , c , $\hbar$) and B depends on the composition only through the mean molecular weight per electron $\mu_e = 1/Y_e$, with Y_e the number of electrons per baryon. This is a purely mechanical equation of state: at fixed composition, pressure depends on density alone, with no temperature dependence. Consequently, once the configuration is restricted to be non-rotating and field-free, the entire stellar structure (mass, radius, density profile) is determined by exactly two physical parameters: the central density ρ_c and μ_e .

which is what makes the self-consistent field (SCF) iteration of Sect. 3 tractable: at each iteration the density is recovered directly from the local enthalpy via the inverse relation $x(H) = \sqrt{(1 + HB/8A)^2 - 1}$, with $H \leq 0$ mapped to $\rho = 0$ (the stellar surface).

Near the surface ($x \ll 1$) the equation of state reduces to $\rho \propto H^{3/2}$, an effective polytropic index $n = 3/2$; in the relativistic core ($x \gg 1$) it approaches $\rho \propto H^3$, $n = 3$ – the marginal polytropic index at which the equilibrium mass becomes independent of central density, the origin of the Chandrasekhar mass limit.

Figure 1 plots Eqs. 1 and 2 and shows where those limits actually apply. The transition is slow: the adiabatic index $\Gamma = d \ln P / d \ln \rho$ takes about four decades in density to fall from $5/3$ to $4/3$, and neither limit is a good description through most of that range. It also places the configuration of Sect. 4. At its central density, $\rho_c = 10^9 \text{ g cm}^{-3}$, the normalized Fermi momentum is $x = 8.0$ and $\Gamma = 1.343$ – within 0.7% of the relativistic limit. The core therefore sits essentially at the marginal index, which matters twice over here: at $n \simeq 3$ the equilibrium is only weakly sensitive to ρ_c and correspondingly delicate, and it is also the regime in which magnetic support translates most directly

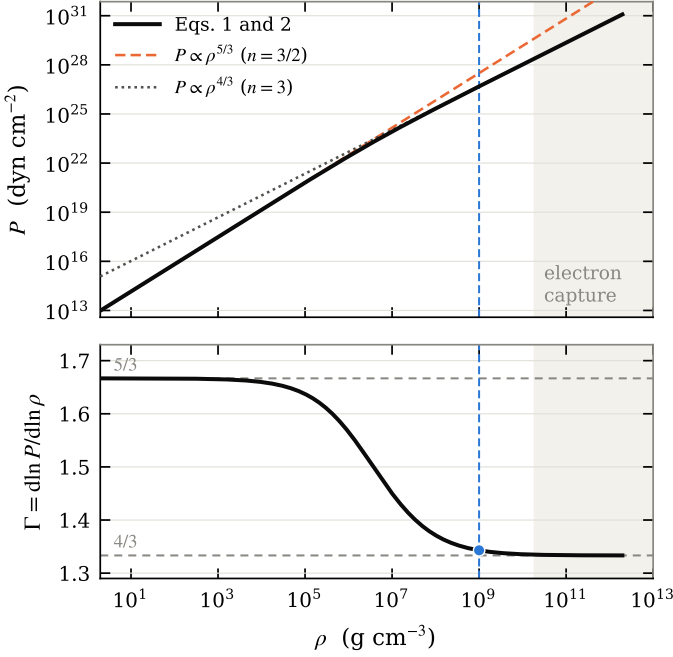


Fig. 1. The equation of state of Eqs. 1 and 2 for $\mu_e = 2$. Upper panel: $P(\rho)$ with the two polytropic limits, each normalized on the curve in the regime where it holds. Lower panel: the adiabatic index, which crosses from $5/3$ to $4/3$ over roughly four decades in density. The dashed vertical line marks the central density of the configuration of Sect. 4, and the point its adiabatic index there. The shaded region lies above the inverse-beta-decay threshold for $\mu_e = 2$, where this equation of state no longer describes a white dwarf.

into mass, since the magnetic energy scales with radius the same way the gravitational and internal energies do.

At sufficiently high density, inverse beta decay (electron capture) changes the composition locally and the constant- μ_e cold equation of state above ceases to describe a real white dwarf. We adopt the tabulated threshold densities of Boshkayev et al. (2013) for this validity limit; all central densities discussed below are kept below this threshold for the assumed composition.

3. Field equations

3.1. The toroidal branch

For a barotropic star the magnetic force must derive from a potential if hydrostatic balance is to close as a Bernoulli integral. For an axisymmetric, purely azimuthal field $\mathbf{B}_t = B_\varphi \hat{\mathbf{e}}_\varphi$ this restricts B_φ to

$$B_\varphi = K \rho^m \varpi^{2m-1}, \quad m \geq 1, \quad (4)$$

with ϖ the cylindrical radius, K the amplitude and m the concentration towards the dense interior. The Lorentz force then follows from a potential, $\mathbf{f}_L/\rho = -\nabla M_{\text{tor}}$ with

$$M_{\text{tor}}(s) = \frac{mK^2}{4\pi(2m-1)} s^{2m-1}, \quad s \equiv \rho\varpi^2, \quad (5)$$

which we verified symbolically for $m = 1, 3/2, 2, 3$. Hydrostatic equilibrium reduces to

$$H + \Phi + M_{\text{tor}}(\rho\varpi^2) = C, \quad (6)$$

solved together with the Poisson equation for Φ and the equation of state of Sect. 2 by the self-consistent field iteration of

Hachisu (1986). Because M_{tor} depends on ρ , the density update is no longer a direct inversion of the equation of state but a per-point root find; the left-hand side is monotonic in ρ , so the root is unique and safely bracketed.

3.2. The poloidal component, and why it is imposed

An axisymmetric poloidal field derives from a flux function $u(\varpi, z) = \varpi A_\varphi$,

$$B_\varpi = -\frac{1}{\varpi} \frac{\partial u}{\partial z}, \quad B_z = \frac{1}{\varpi} \frac{\partial u}{\partial \varpi}, \quad (7)$$

with u obeying the Grad-Shafranov equation (Grad & Rubin 1958; Shafranov 1966) sourced by the current distribution, which we take linear in the flux.

A single self-consistent solve carrying both components is not available in this formulation, and the obstruction is not a numerical one. In barotropy the toroidal function must depend on the flux alone, $\beta = \beta(u)$; since there is no current outside the star, B_φ must vanish on every flux surface that reaches the surface, so the toroidal field is confined to the region of closed poloidal lines. The energy ratio reachable that way is a few percent, far from the toroidal-dominated configuration required below. We therefore solve the toroidal branch self-consistently for the structure, and impose the poloidal field on the converged density.

That imposition has a price, and Sect. 4.4 measures it: the imposed field's energy is precisely the term left unbalanced in the virial, so imposing a poloidal component carrying a fraction x of the binding energy costs a virial error of about x . The poloidal component adopted here is small enough that this cost stays below the acceptance threshold, which is what makes the configuration certifiable at all.

Purely toroidal and purely poloidal equilibria are each subject to a non-axisymmetric instability of azimuthal wavenumber $m = 1$ (Tayler 1973; Markey & Tayler 1973), which an axisymmetric formulation cannot test for. Sect. 5 returns to this.

4. Results

4.1. A certified $2 M_\odot$ configuration

Table 1 reports the configuration this paper is about: a non-rotating white dwarf of $2.007 M_\odot$ at $\rho_c = 10^9 \text{ g cm}^{-3}$, carrying a toroidal field with $E_{\text{tor}}/|W| = 0.2026$ and a poloidal component imposed on top.

The mass gain is the whole point of the toroidal field: the same central density and composition without a field give $1.347 M_\odot$ at a radius of 2457 km, so the field is carrying +49% in mass and inflating the star by a factor 2.1 in equatorial radius. A $2 M_\odot$ white dwarf of radius 5200 km is a large object by the standards of its mass – unsupported, a star this massive would be far more compact, and the size is the visible form of the support the field is providing. That gain is available at a central density well below the inverse-beta-decay threshold for $\mu_e = 2$ ($1.94 \times 10^{10} \text{ g cm}^{-3}$, Sect. 2), so the configuration is a white dwarf in the sense the equation of state can still describe.

The shape of the star is not what $R_{\text{pol}}/R_{\text{eq}}$ suggests. That ratio is 1.10 and describes only the surface where the enthalpy vanishes; the mass distribution is far more extreme. Measured at matched spherical radius, the density on the rotation axis exceeds the density at the equator by a factor 11.5 at $0.2 R_{\text{eq}}$ and 23.4 at $0.5 R_{\text{eq}}$. Along the equator the density falls by a factor 230 from the centre to half a radius; along the axis it falls by 10.

Table 1. The adopted configuration. $\mu_e = 2$, $m = 1$, no rotation.

ρ_c (g cm ⁻³)	1.0×10^9
M ($M_\odot$)	2.007
R_{eq} (km)	5207
R_{pol} (km)	5736
$R_{\text{pol}}/R_{\text{eq}}$	1.102
$\rho_{\text{axis}}/\rho_{\text{eq}}$ at $0.2 R_{\text{eq}}$	11.5
$\rho_{\text{axis}}/\rho_{\text{eq}}$ at $0.5 R_{\text{eq}}$	23.4
K	3.245×10^{-3}
$E_{\text{tor}}/ W $	0.2026
$B_{\varphi, \text{max}}$ (G)	4.275×10^{13}
B_{pole} (G)	5.91×10^{10}
B_t/B_p (energy)	884
B_t/B_p (amplitude)	26.2
$E_{\text{pol}}/E_{\text{mag}}$	1.1×10^{-3}
VE	9.9×10^{-5}
f_{pol}	0.142

The configuration is a dense axial column inside an evacuated equatorial region.

That follows directly from the field law. The magnetic pressure of $B_\varphi = K\rho\varpi$ goes as $\rho^2\varpi^2$, so it is largest exactly where the density and the cylindrical radius are both large, and it drives matter out of the equatorial mid-region and onto the axis. The effect strengthens with K : across configurations of 1.50, 1.70, 1.85 and 2.01 $M_\odot$ at this central density, the ratio at $0.5 R_{\text{eq}}$ runs 3.0, 8.0, 14.2 and 23.4, while $R_{\text{pol}}/R_{\text{eq}}$ moves only from 1.037 to 1.102. The surface ratio is a poor proxy for the deformation and we quote both.

The field is toroidal-dominated by a wide margin, $B_t/B_p = 884$ in energy and 26 in amplitude. The two differ because energy goes as the square and the components occupy different volumes: the toroidal field is confined to the interior while the poloidal one extends through the star and beyond it. Calling this a mixed field is accurate but should not be read as balanced – the poloidal component carries a tenth of a percent of the magnetic energy. What it does carry is the exterior structure, which the toroidal component cannot have at all: B_φ vanishes with the density, so a purely toroidal star has no field outside it and no dipole for magnetic braking to act on. The poloidal component supplies a surface dipole of 5.9×10^{10} G.

One number worth stating next to the field strength. The critical field at which Landau quantization changes the electron equation of state is $B_c = m_e^2 c^3 / e \hbar = 4.414 \times 10^{13}$ G, and the peak field here is $0.97 B_c$ – just below it. The equation of state used has no field dependence, so a configuration above B_c would be inconsistent with its own microphysics; this one sits at the edge of that regime rather than inside it.

4.2. The field inside the star

Figures 2 and 3 show the field the previous section quantifies, first the poloidal component alone and then the full field. Both take a form dictated by the axisymmetry. A poloidal line has no azimuthal component, so it never leaves its meridional plane, and there $\mathbf{B}_p$ is perpendicular to ∇u : the poloidal lines are the contours of the flux function, exactly, and Fig. 2 draws them as such rather than integrating them. Each is a planar closed loop about the O-point, and the three-dimensional poloidal field is those loops repeated at every azimuth. The lines of the full field are integrated in cylindrical coordinates for the same reason – it keeps them on their flux surfaces by construction, and the drift of u along a traced line then measures the integration error rather

than hiding it. Over the lines drawn it stays between 10^{-4} and 6×10^{-4} .

The two components are complementary in where they live. B_φ vanishes on the axis, where $\varpi \rightarrow 0$, and at the surface, where the density does; $|\mathbf{B}_p|$ does the opposite, peaking on the axis near the centre at 1.5×10^{12} G against 1.2×10^{10} G at half the stellar radius in the equatorial plane, a hundredfold contrast. The poloidal field also reaches outside the star, where ρ and hence B_φ are zero, so the exterior field is purely poloidal and roughly dipolar: u falls off as $1/r$ beyond the surface, and it is this component alone that a distant observer could detect.

Which loops close inside the star is fixed by the flux at the stellar surface. Along the equator u rises from zero on the rotation axis to its maximum u_{axis} at the O-point of the poloidal field, at $\varpi = 3.06 \times 10^8$ cm, and then falls to $0.775 u_{\text{axis}}$ at the surface – it does not return to zero. That value is a separatrix. Surfaces with $u > 0.775 u_{\text{axis}}$ close inside the star and carry the toroidal field; surfaces below it intersect the stellar surface, and a line on one of them leaves the star, where ρ and therefore B_φ vanish and the line stops winding altogether. The closed region is the twisted torus proper. Fig. 2 shows the separatrix directly: its inner three loops close inside the star and its outer two bulge through the surface and close outside it, and Fig. 3 winds its helices on the inner three. They are seeded by flux level rather than by radius, since u is what labels the surfaces: equally spaced radii land on unequally spaced surfaces and can straddle the separatrix, putting two structurally different objects in one picture.

Followed for one meridional circuit each, those lines wind 17.5, 18.3 and 19.4 times around the axis, the winding increasing outward through the closed region. This is the twisted-torus geometry with the twist taken to an extreme: the poloidal circuit is barely traced out before the line has wrapped the axis a score of times.

These lines do not close on themselves, and the winding number is what decides that. A line confined to a flux surface returns to its starting meridional plane after each circuit, but at an azimuth advanced by the winding number; it closes only if that number is rational. Followed for eight circuits, the line drawn from mid-radius has a winding of 17.476, constant to three decimals, so each circuit returns it a further 0.476 of a turn around the axis and it never lands back where it started: its closest approach over those eight circuits is $0.15 R_\star$, after two circuits, where twice the fractional part is nearest an integer. The line is quasi-periodic, covering its torus densely rather than retracing a closed curve, and only the measure-zero set of surfaces with rational winding carries closed field lines.

That is a middle case worth naming, because the two extremes both occur in this problem. A purely toroidal field has closed lines trivially – circles about the axis, since nothing drives a poloidal drift. A field with no symmetry has chaotic lines that fill a volume. An axisymmetric mixed field lies between: the symmetry confines each line to a two-dimensional surface, but nothing confines it to a curve on that surface.

The winding is a property of each flux surface and is not the amplitude ratio of Table 1 restated. That ratio is $\max |B_\varphi| / \max |\mathbf{B}_p|$, with the two maxima taken at different places in the star, so it is a global number rather than a local one. Along the lines drawn here the local $|B_\varphi|/|\mathbf{B}_p|$ varies by nearly an order of magnitude from surface to surface, while the winding number changes by about a tenth. The two quantities answer different questions.

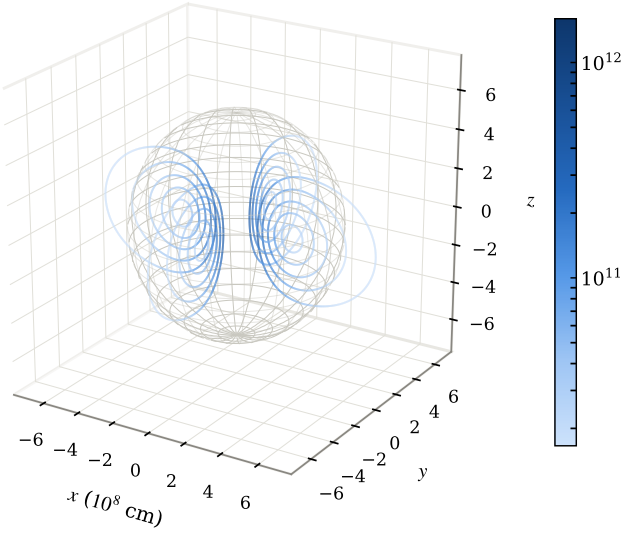


Fig. 2. The poloidal component alone, in three dimensions, coloured by $|B_p|$. A poloidal line stays in one meridional plane and is exactly a contour of the flux function there, so each is a planar closed loop about the O-point; loops are drawn at $u/u_{\text{axis}} = 0.97, 0.90, 0.80, 0.68$ and 0.55 , and repeated at four azimuths. The first three close inside the star and are the surfaces Fig. 3 winds its lines on; the last two cross the surface and close outside, where the density and hence B_ϕ vanish and the field is purely poloidal. The wireframe is the stellar surface itself, prolate as a toroidal field makes it, not a sphere.

4.3. Convergence

Figure 4 shows the configuration under mesh refinement at fixed domain, halving Δr four times. The mass converges monotonically with shrinking increments – 0.15%, 0.04%, 0.01% – to $2.007 M_\odot$, and $E_{\text{tor}}/|W|$ is converged to four significant figures at every resolution beyond the coarsest.

The virial error stays below the 10^{-3} acceptance threshold at all four resolutions, but it is not monotonic: it dips at the second resolution and settles near 10^{-4} . We report this rather than smoothing it over, because it means the virial error at a single resolution is not an error estimate for this configuration; the mass convergence is. The same non-monotonicity appears at larger K in this branch, where it does cross the threshold; here it does not.

The domain was sized by the criterion that the star’s polar radius clear the boundary, $f_{\text{pol}} = 0.142$ against the 0.2 limit adopted for this branch, and the configuration is reached by continuation in K from the field-free solution, since direct solves at this amplitude do not converge.

4.4. What the poloidal component costs

The poloidal field is imposed rather than solved for, so it leaves the pair out of virial balance by its own energy. Figure 5 measures that directly: as the poloidal amplitude is raised, the virial error of the combined configuration tracks $E_{\text{pol}}/|W|$ almost one for one, and the pair leaves the certified band once the exterior dipole exceeds roughly 10^{11} G.

This sets the configuration adopted above. At $B_t/B_p = 884$ the virial error is 9.9×10^{-5} , an order of magnitude inside the threshold, and the exterior dipole is 5.9×10^{10} G. Reaching a more balanced field is possible but not certifiable by the standard applied here: at $B_t/B_p = 8.8$ the virial error is 2.3×10^{-2} , twenty times the threshold. A configuration in that range would have to be justified dynamically – loaded into an evolutionary

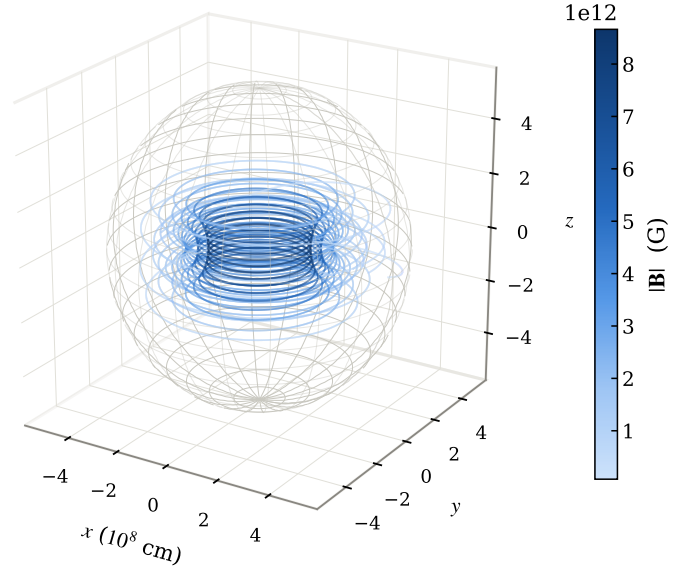


Fig. 3. Field lines of the same configuration in three dimensions, coloured by field strength, with the stellar surface as the faint wireframe – the true surface of revolution, as in Fig. 2, drawn whole so its prolateness and the depth of the lines inside it can both be judged. Each line is drawn for exactly one meridional circuit. The three are seeded at $u/u_{\text{axis}} = 0.97, 0.90$ and 0.80 , all inside the closed region $u > 0.775 u_{\text{axis}}$, so each lies on a torus contained in the star; they wind 17.5, 18.3 and 19.4 times about the axis over that circuit, the winding rising outward. The lines do not close on themselves: the winding number is not an integer, so each circuit returns a line to its starting meridional plane at a different azimuth.

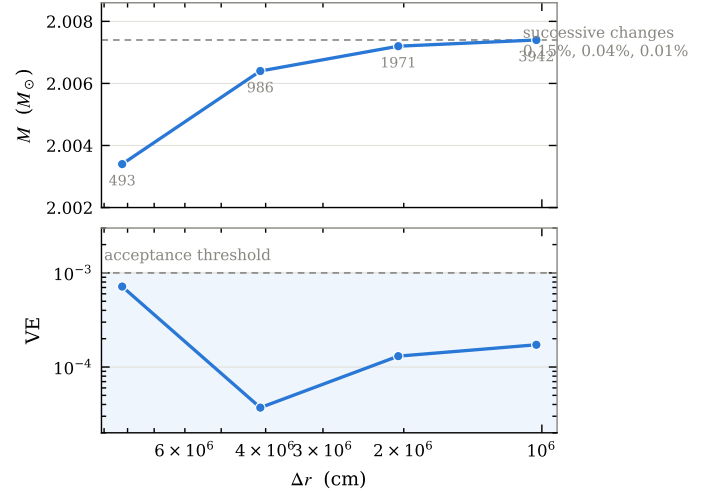


Fig. 4. Mesh convergence of the adopted configuration at fixed domain, labelled by radial resolution. Upper panel: the mass, converging monotonically to $2.007 M_\odot$; the dashed line is the finest value. Lower panel: the virial error, which stays inside the acceptance band (shaded) at every resolution without being monotonic.

calculation and allowed to relax – rather than presented as an equilibrium.

5. Limitations

The configuration does not survive dynamically. The equilibrium above is certified to satisfy hydrostatic and virial balance and nothing more, and a three-dimensional magnetohydrodynamic evolution of it does not hold. Exported as a vector poten-

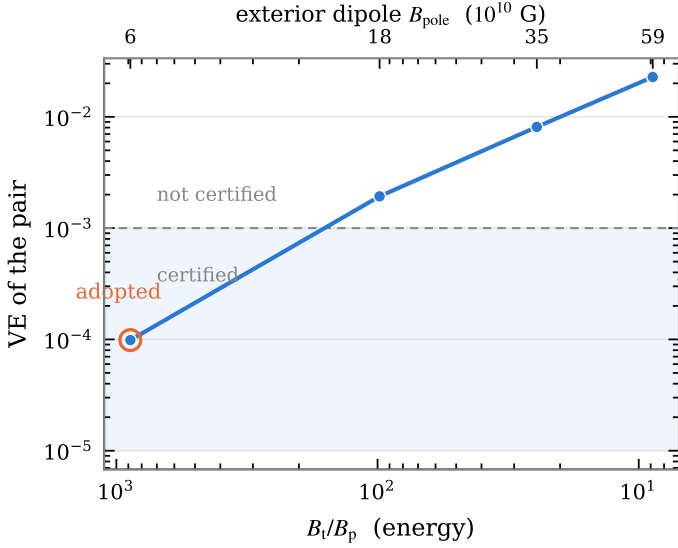


Fig. 5. The cost of the poloidal component. Moving left to right, the imposed poloidal amplitude rises, the field becomes less toroidal-dominated, and the exterior dipole (upper axis) grows; the virial error of the pair rises with it and leaves the certified band (shaded). The circled point is the configuration of Table 1.

tial and evolved in CASTRO, it undergoes a large radial expansion, rebounds, and runs away in central density, reaching $10^{10} \text{ g cm}^{-3}$ and terminating at $t \simeq 2.5 \text{ s}$, about three dynamical times.

The collapse is not an artefact of the transfer, as far as we have been able to test. It is unchanged by resolution: at 64^3 , 96^3 and 128^3 the peak field is reconstructed to 84.5%, 91.4% and 95.6% of its correct value, taking the local magnetic force from 71% to 91%, and the survival time moves only from 2.18 to 2.70 s. Mass is conserved to 0.02%, the centre of mass does not drift, and the density maximum stays at the origin throughout. A field-free equilibrium mapped and evolved the same way is bounded, its central density oscillating between 0.86 and 0.96 of its initial value.

It is also insensitive to the field strength, which is the informative part. Across 1.50 to $2.01 M_\odot$, spanning a factor 2.8 in $E_{\text{tor}}/|W|$ and a factor 7 in the equatorial evacuation described in Sect. 4, the termination time moves by 12%. What is invariant across those configurations is the axial column: the density on the axis at $0.2 R_{\text{eq}}$ agrees to 1.9% among them, because $B_\varphi = K\rho\varpi$ vanishes on the axis and the toroidal field does not reach the material that runs away.

The instability is therefore radial and central, and the non-axisymmetric Tayler mode (Tayler 1973; Markey & Tayler 1973) – which a field this toroidal is the classic candidate for, and which motivated the evolution in the first place – is never reached. Whether some other field geometry supports $2 M_\odot$ is not addressed by these runs; what they establish is that the self-consistent toroidal branch with $B_\varphi = K\rho\varpi$ at $E_{\text{tor}}/|W| \sim 0.1\text{--}0.2$ does not.

The poloidal component is not self-consistent. It is solved on the converged toroidal density and imposed, for the reason given in Sect. 3.2: barotropy does not deliver a strongly mixed equilibrium. The resulting virial error is small enough to certify, but the configuration is not a solution of a single set of equations carrying both components.

The equation of state does not respond to the field. At $0.97 B_c$ the peak field is at the edge of the regime where Landau quantization modifies the degenerate electron gas, and the inverse-beta-decay threshold used to bound ρ_c would itself shift with field (Das & Mukhopadhyay 2013). Neither effect is included. Staying below B_c keeps the configuration self-consistent in this respect, but only barely, and it forecloses raising K further at this central density.

Barotropy. The star is described by a cold, temperature-independent equation of state, so it has no entropy stratification and no buoyancy restoring force. Stable magnetic equilibria in stars are usually argued to rely on stable stratification, which this model does not have; whether that is a limitation of the model or a statement about degenerate stars is beyond what these calculations can settle.

Acknowledgements. TBD.

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